

FEASIBILITY STUDY OF HYDROKINETIC POWER FOR ENERGY ACCESS IN RURAL SOUTH AFRICA

Kanzumba Kusakana, Herman Vermaak
Department of Electrical Engineering and Computer Systems
Central University of Technology, Free State
Bloemfontein, South Africa
kkusakana@cut.ac.za, hvermaak@cut.ac.za

ABSTRACT

Renewable energy advocates, the South African Government and investors are more and more becoming aware of stream, river currents and their enormous associated energy potential. Since hydrokinetic power generation (HKP) relies basically on the extraction of energy from the natural velocity of free flowing water, this power system can be categorised as sources of flowing water with minimal civil infrastructure, environmental impacts and costs.

This study investigates the possibility of using and developing hydrokinetic power to extend the reliable, affordable and sustainable electricity supplies for rural, remote and isolated loads in rural South Africa where reasonable water resource is available. Simulations are performed using the Hybrid Optimisation Model for Electric Renewable (HOMER) and the results are compared to those from other supply options such as standalone Photovoltaic system (PV), diesel generator (DG) and grid extension. Finally the paper points out some major challenges that are facing the development of this technology in South Africa.

KEY WORDS

Hydrokinetic power, renewable energy, rural electrification, South Africa

1. Introduction

South Africa is endowed with abundant renewable energy resources that can be used optimally to help facing the challenges of global warming, reduce green house gases emissions resulting from the extensive use of fossil fuel as primary resource of electric energy and to have an energy security through diversification of supply [1]. It is in this context that the South African Government is giving a push to renewable energy and integrates it into the mainstream energy economy. To reach this goal, South African Government is setting a target 10 000 GWh renewable energy contribution to be produced mainly from biomass, wind, solar and small-scale hydropower by 2013 [2].

Hydrokinetic power generation is a category of hydropower energy that extracts kinetic energy from

flowing water rather than potential energy from water fall. Hydrokinetic power systems avoid many of the challenges which are coming across with traditional hydropower, such as high civil infrastructure costs, and the need of acceptable water head [3]. They have simple design and can be easily installed and maintained by local population at low cost if installed in remote and rural areas. Another advantage is that hydrokinetic can be easily installed in free-flowing rivers or streams to enhance energy extraction, these make hydrokinetic far more competitive compared to small traditional hydropower even though they can extract almost the same amount of energy.

Approximately 6000 to 8000 potential sites for traditional micro hydropower applications are situated mainly in Eastern Cape and KwaZulu-Natal provinces [4]. Due to the simplicity of the hydrokinetic power design, there are theoretically huge numbers of potential sites as compared to small hydropower generation. The cost of energy extracted from Hydrokinetic is lower than the one of small hydropower. Hydrokinetic technology is more economical compared to solar power system; it is thus a better candidate for South African rural electrification programmes where water resource is available.

This study investigate the possibility of using and developing hydrokinetic power to extend the reliable, affordable and sustainable electricity supplies for rural and remote loads in South Africa where reasonable water resource is available. For this purpose, we have selected a potential site from which we have acquired data such as water flow and energy demand needed as input to the HOMER program. The simulation results of the proposed hydrokinetic system are compared to those from other power supply options such as standalone PV, diesel generator and grid extension line to find the optimal and most suited option to supply the rural and isolated load.

2. Hydropower Situation in South Africa

2.1 Hydropower Potential

By international standards, the extensive development of hydropower for electricity generation has not yet been considered seriously in South Africa. No significant

development of hydropower in the country has been noted for 30 years, except the new small scale installation of 7MW capacity commissioned at the Sol Plaatjie Municipality Free State province. At the present the overall hydroelectricity generation capacity represents only about 5% of present total 45 500MW installed generation capacity [5]. Table 1 below gives a summary of the hydropower potential is South Africa

Table 1
Hydropower potential in South Africa

Size	Type	Installed capacity (MW)	Estimated potential (MW)
Macro hydropower (Larger than 10MW)	(i) Imported	1 450	36 400
	(ii) Pumped storage for peak supply	1 580	10 400
	(iii) Diversion fed	-	5 200
	(iv) Dam storage regulated head	662	1 520
	(v) Run of river	-	270
Small hydropower (from a few kW to 10 MW)	As above (iv) and (v)	29.4	113
	Water transfer	0.6	38
	Refurbishment of existing plants	8.0	16
	Gravity water carrier	0.3	80
Sub-total for all types		3 730.3	53 837
Excluding imported from abroad		2 280.3	17 437
Excluding pump storages using coal based energy		700.3	7 237
Total “green” hydro energy potential available within the border of South Africa			7 237

2.2 Where to Look for Hydroelectricity in South Africa

The rural communities in the Eastern Cape, Mpumalanga and KwaZulu-Natal provinces have access to water resources with good hydropower potential [6]. The development of small-scale traditional hydroelectric installation particularly for the commercial and domestic consumption should be strongly promoted and supported. Communities with hydropower potential and interest in developing hydro-electricity need a wide professional support since any new hydropower installation is costly and requires technical and operational inputs from civil, mechanical and electrical professionals. The figure below

shows the areas where potential site for development of micro hydropower as well as the location where small hydropower plants have already been implemented [7].

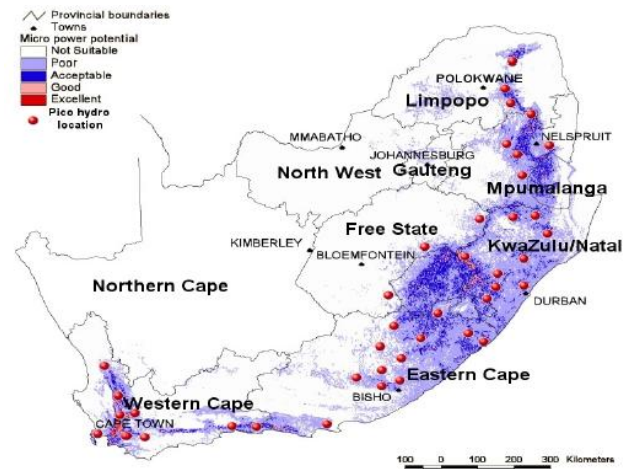


Figure 1: Small scale hydropower distribution in South Africa

We have to notice that the figure1 and table 1 above do not take into consideration the energy potentially available from hydrokinetic which can represent a potential source of electric power even greater than the one from Micro and Pico hydropower plants. The ideal location for a hydrokinetic turbine is to be located in deep strong flowing rivers or immediately downstream from an existing conventional hydropower plant where electric transmission wires and interconnection facilities are located, and also where the energy remaining in the water current existing from the turbines in the dam can be reused. Theoretically, a greater number of potential sites to implement hydrokinetic power can be identified compare the traditional small-scale hydropower.

3. Hydrokinetic

3.1 Technology

Hydrokinetic was originally developed to surmount the numberless of problems associated with dams throughout the world. This system is erected into the river or stream which results in the following advantages compared to the traditional hydropower:

- No dam,
- No destruction of nearby land,
- No change in the river flow regime,
- Reduction of flora and fauna destruction.

The terms hydrokinetic encapsulate both tidal and river applications. Within the context of this paper, the focus is on river application, since it is suitable for energy generation at remote and isolated locations.

Most of the principals of this type of turbine are based upon wind turbines, as they work in a similar way but with the possibility of having close to 1000 time more energy from the hydrokinetic compared to traditional hydropower [8]. The power available (P_a) in watts can be worked out using the following equation.

$$P_a = \frac{1}{2} \times A \times \rho \times V^3 \times C_p \quad (1)$$

A = area in metres squared (m^2)

ρ = density of water (1000 kg/m^3)

V = velocity of water (m/s)

C_p = the power coefficient = $16/27 = 0.592$ (theoretical maximum power available)

The theoretical maximum power available from the river is expressed by the equation above using a power coefficient of 0.592 or 59% efficiency. But a small-scale river turbine has its own losses which will reduce the power coefficient to around 0.25.

From equation 1 above, it is noticeable that the power increases in a cubed relationship to the velocity of the flow of water past the turbine. Therefore it is important to find the best flow to get the best power output.

3.2 The Turbine

In order to calculate the diameter of the rotor, following formula is applied:

$$P = \frac{1}{2} \times \rho \times A \times V^3 \times C_p \times \eta \quad (2)$$

Where:

$$A = (\pi \times \frac{d^2}{4})$$

η = efficiency of the generator

$$\lambda = \frac{U}{V_D} = \pi \times N \times \frac{d}{60} \times V_D$$

$$N = (60 \times \rho \times \frac{V_D}{\pi} \times d)$$

N = velocity of turbine rotor (r.p.m)

λ = tip speed ratio

U = tangential velocity at the tip of the blade (m/s)

V_D = design velocity (m/s)

3.3 Generator

In order to reduce costs, and to be able to rely on locally-made technology, Practical Action began by working on the development of a permanent magnet generator. The magnets allowed the speed of generation to be reduced, and lowered the cost of the equipment, which itself could be adapted to be a river turbine rotor, and ultimately, tested and built.

4. System Design and Simulation

The HOMER simulation program has been chosen as a tool for system design. HOMER was selected due to its capability to evaluate the best option by harnessing energy from a single or combination of various energy resources [9]. It is an economic model that provides rational selection of the most cost effective option [10]. Furthermore, its hourly energy flow approach offers a comprehensive analysis of the system performance throughout a year. A potential site has been selected from which the load energy demand, the renewable energy resources, as well as the cost of the supply options (Hydrokinetic, solar PV, wind, diesel generator and grid extension) have been used as input HOMER.

4.1. System Input

4.1.1 Load

Table 1 gives domestic appliances, power demand and running times for an average typical household in rural South Africa [11].

Table 1
Domestic power demand estimation

Equipment	Amount	Power demand (kW)	Running time (hour)	Energy (kWh/d)
Light	5	0.006	6	0.18
Radio	1	0.020	5	0.1
T.V.	1	0.07	5	0.35
Iron	1	1	0.1	0.1
Kettle	1	1.5	0.05	0.075
Fridge	1	0.12	24	2.88
Mobile phone charger	3	0.004	1	0.012

The daily energy consumption (9.5kWh) is the sum of the daily consumption of the equipments, the peak load at 3.4 kW and a 0.12 load factor.

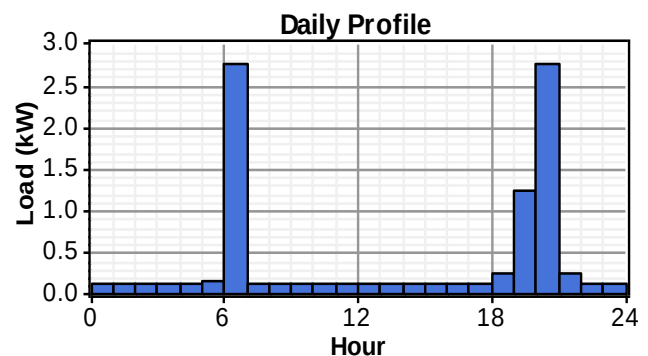


Figure 2: Typical rural daily load profile

4.1.2 Resources

- **Hydro Resources**

The summary of the physical characteristics of the selected river is shown in Table 2. A correction factor of 0.8 has been applied to the measured values to accommodate friction effects along the bottom and sides of the river on the current velocity [12].

Table 2
Site water velocity

Month	Velocity (m/s)
January	5.31
February	7.25
March	6.09
April	1.81
May	2.67
June	2.18
July	1.84
August	1.54
September	1.41
October	1.69
November	2.83
December	5.27
Average	3.32

The theoretical potential power available from the site can be found with the help of equation (1) using the following river characteristics:

- Water velocity in the worst month: 1.41 m/s
- Viable depth: 1,8m
- Width: 5,2m
- Cross sectional area: 9.36m²
- $P_a = 1,075$ kW

With reasonable sizing of the battery storage system, this available power can cover the load energy requirement without interruption.

Unfortunately, HOMER is not equipped with a hydrokinetic power module considered in this study. Consequently, instead of using its hydro module, the wind turbine components has been used with hydrokinetic input rather than wind-related information. This approach was considered because wind turbines share some similarities with hydrokinetic turbines which are commonly referred to as ‘underwater wind turbines’. Thus, the wind turbine power-curve has been replaced with the power curve of the selected hydrokinetic turbine by altering the wind speed information with the river current velocity [13].

- **Solar resources**

The selected site has good solar resources but very poor in wind, so only the solar system and the stand alone diesel generator can be compared to the hydrokinetic while supplying the same load to find out which one is the best renewable supply option for the site.

Table 3
Site solar resources

Month	Clearness index	Daily radiation (kWh/m ² /d)
January	0.627	7.404
February	0.646	7.178
March	0.639	6.274
April	0.638	5.200
May	0.698	4.663
June	0.758	4.522
July	0.743	4.663
August	0.641	4.804
September	0.690	6.302
October	0.607	6.443
November	0.561	6.500
December	0.553	6.613
Average	0.638	5.873

4.2 Components Information

4.2.1 Hydrokinetic Power

As mentioned above the hydrokinetic turbine information was inserted into the Wind Turbine Input window. The Darrieus Hydrokinetic Turbines (DHT) developed by Alternative Hydro Solutions in Canada has been chosen because of its simple structure and its ability to generate a relatively high power output from low to medium flow velocities. Figure 3 is the power curve of the turbine based on information from the turbine’s manufacturer. The turbine’s rated power is 1 kW at 1.4 m/s current velocity. Since the information about the power outputs at the flow speed above 1.5 m/s flow was not available, it has been assumed that above 1.5 m/s, there are no increases in power output.

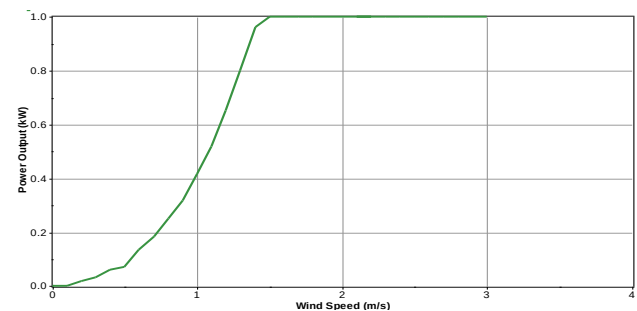


Figure 3: Hydrokinetic power curve

The investments as well as the replacement costs of the 1kW hydrokinetic turbine are \$9170 respectively, the operation and maintenance cost is \$20/year; the system lifetime span is 25 years [14].

4.2.2 Photovoltaic System

The actual price of the PV module is set at 3.59 \$/W in USA, considering the transport and other unpredictable costs. The price of PV is set to 4500 \$/kW with the

replacement cost of \$4100. The cost of the inverter is set 800 \$/kW [15]. The operation and maintenance costs of the Photovoltaic module and of the inverter are estimated at 105 and 10 \$/yr respectively [16]. The price of the deep cycle battery is \$215 with a replacement cost of \$215, the operation and maintenance is \$5 [17]. The life time of the Photovoltaic array and battery is taken as 20 year.

4.2.3 Diesel Generator

Given that the peak power demand is 3.4kW, the diesel generator cost is taken as \$2830 [18]. The operating and maintenance costs are 0.5\$/h and the fuel consumption (0.55litres/kWh). In the South Africa the price of the diesel and lubricant is 1.2\$/ℓ and 1.30\$/ℓ respectively.

We have also to take in account the international carbon emission penalty of 2.25 \$/t.

5. Simulation Results and Discussion

HOMER simulates system configurations with all of the combinations of components that were specified in the component input. It discards from the results, all non-feasible system configurations, which are those that do not adequately meet the load, given either the available resource or constraints that were specified.

The architectures and costs of different supply options found feasible by Homer are presented below in the tables 4 and 5 respectively:

Table 4
Systems architecture

Systems	HKP	PV	DG
Size (kW)	1	6	3.4
Number of Battery	7	12	0
Inverter (kW)	3.5	3.5	0
Rectifier (kW)	3.5	3.5	0

Table 5
Simulation results summary

Costs	HKP	PV	DG
Capital (\$)	11 475	7 880	2 830
Replacement (\$)	4 697	9 344	20,300
O&M (\$)	831	2 337	55 991
Fuel (\$)	0	0	75 971
Salvage (\$)	-290	-967	-264
Total NPC (\$)	16 713	18 495	154 829
COE (\$/kWh)	0.387	0.416	3.475
Grid extension distance (km)	0.911	1.07	13.1

The technical and economical results obtained by Homer are displayed on the table 5. Form this table we can notice that based on the Net Present Cost (NPC), Cost of Energy (COE) and the breakeven grid extension distance that the micro hydropower is the best option to supply the load with electricity.

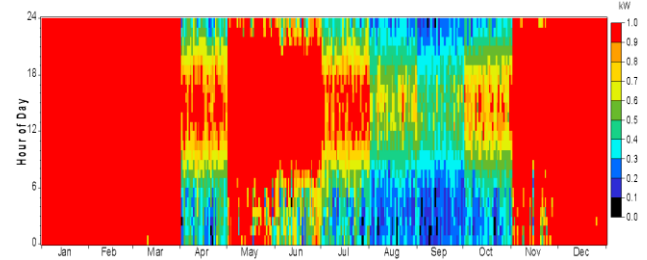


Figure 3: Micro hydro output

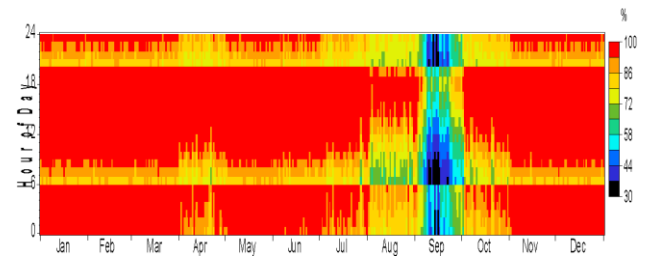


Figure 4: Battery state of charge

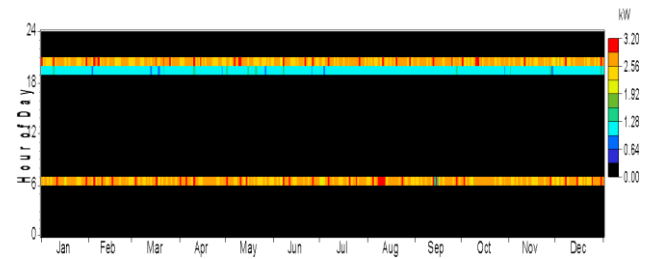


Figure 5: Inverter output

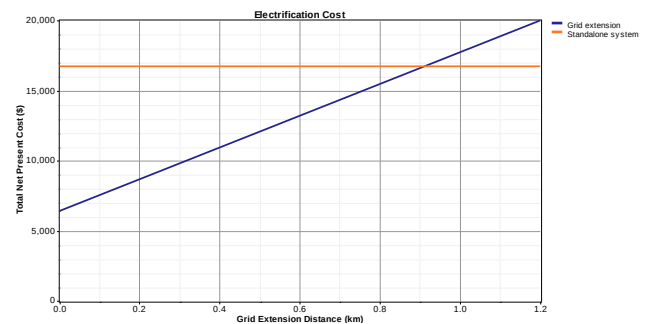


Figure 6: Grid extension distance

Figure 3 show the average monthly hydrokinetic output power. During the month of September, due to insufficient water resource the hydrokinetic plan gives an average of 1 kW which is its minimum output.

Figures 4 and 5 show the converter usage and the battery state of charging. It can easily be seen that from April to November the battery system is charged during

off-peak times and used during peak power demand times occurring in the mornings and evenings.

Figure 5 show that the power from the battery is used only during morning and evening peak times to compensate the hydrokinetic deficit in power supply.

Figure 6 gives the breakeven grid extension distance at 0.911 km. This means that the total cost of implementing the micro hydro project for 25 years will be equivalent to the cost of installing a grid extension line of 0.911 km.

6. Conclusion and Recommendations

This paper aimed to investigate the possibility of using and developing hydrokinetic power suitable to supply electricity to rural and isolated loads in South Africa where reasonable water resource is available

The proposed hydrokinetic system is sized to meet the load energy requirement during the worst months. Simulations of the hydrokinetic power have been performed with HOMER software. The results have been compared with those of a diesel generator and PV while they are supplying the same load. The hydrokinetic system (composed of 1kW turbine, 3.5kW inverter, and 7 batteries) has an initial capital cost of \$11 475, a Net Present Cost of \$16 713, operation cost of \$831 per year and energy production cost of 0.387 \$/kWh. This makes hydrokinetic the best supply option compared to the PV and diesel generator. Apart for being very cheap to operate and maintain, the hydrokinetic system contributes to the reduction of the CO₂ and green house gases in the atmosphere.

The results of this study have led to the following further study recommendations:

- Identify sites and assess potential energy available,
- Develop policies supporting the development of hydrokinetic power in South Africa.

References

[1] Department of Minerals and Energy Affairs see SOUTH AFRICA. Department of Mineral and Energy Affairs. "Implementation of Strategy for Renewable Energy in South Africa". Draft 2. 2000. Pretoria: Government Printer.

[2] DME, 2003 "White Paper on the Energy Policy of the Republic of South Africa", Department of Minerals and Energy, Pretoria, November 2003.

[3] Wang, L., Infante, D., Lyons, J., Stewart, J. and Cooper, A. "Effects of dams in river networks on fish assemblages in non-impoundment sections of rivers in Michigan and Wisconsin", USA, River. Res. Applic., 2010. [Online] Available from: <http://www.interscience.wiley.com> [Accessed 19 September 2011].

[4] Department of Minerals and Energy Affairs (DME). 2001. "Eskom and Council for Scientific and Industrial

Research (CSIR) Renewable Energy resource GIS Database".

[5] "Baseline study on hydropower in South Africa". Department of Minerals and Energy, Pretoria 2002.

[6] Hearn IR, "Energy Sources for Remote Area Power Supply - A Review and Assessment ", in Proceedings of the conference E-I 92 Electricity beyond the Grid, 1992. 60-75.

[7] B. Barta. "Status of the small-scale hydroelectric (SSHE) development in South Africa", March 2010.

[8] M.A. Arango. "Resource Assessment and Feasibility Study for Use of Hydrokinetic Turbines in the Tailwaters of the Priest Rapids Project". MSc dissertation, department of Mechanical Engineering, University of Washington, 2011.

[9] National Renewable Energy Laboratory (NREL), Hybrid Optimization Model for Electric Renewable (HOMER), 2005. [Online]. Available from: www.nrel.gov/international/tools/tools.html. [Accessed: 29 November 2011].

[10] T. Givler and P. Lilienthal, "Using HOMER software, NREL micropower optimization model, to explore the role of Gen-sets in small solar powersystem: Case study Srilanka," 2005. [Online] Available: <http://www.scribd.com/doc/6610003/Homer-Case-Study-Sri-Lanka?autodown=pdf> [Accessed: 12 April 2011].

[11] K. Kusakana, J.L. Munda, A.A. Jimoh, "Economic and Environmental Analysis of Micro Hydropower System for Rural Power Supply" PECon 2008, The 2nd IEEE Power and Energy conference, Johor Bahru MALAYSIA, 1-3 December 2008, 41-44.

[12] J. Twidell and T. Weir, "Renewable Energy Resources 2nd edition". New York: Taylor & Francis, 2006.

[13] Khan MJ, Iqbal MT, Quaicoe JE. "Design considerations of a straight bladed darrieus rotor for river current turbines", Industrial Electronics, 2006 IEEE International Symposium, Montreal, 9-13 July 2006, 1750-1755.

[14] Kunaifi S. "Options for the Electrification of Rural Villages in the Province of Riau, Indonesia", MSc dissertation, School of Engineering and Energy Murdoch University, Perth Australia, 2009

[15] 125 Watts and Higher Module Index, Retail Price Per Watt Peak. [Online], Available from: <http://www.solarbuzz.com/Moduleprices.htm> [Accessed: 07 November 2011].

[16] Ahmed S. et al, "Optimal Sizing of a Hybrid System of Renewable Energy for a Reliable Load Supply without Interruption" European Journal of Scientific Research , [17] Vol.45 No.4 (2010), pp.620-629

Trojan battery L16p price, [Online], available from <http://www.solar-review.com/reviews/read-review.cfm?id/8/> [Accessed: 07 November 2011].

[18] Price of 12kW diesel generator, [Online] available from: <http://www.peakpowertools.com/Standby-Generator-Diesel-Pramac-12-kW-Watt-p/gprobw15p.htm> [Accessed: 07 November 2011].